

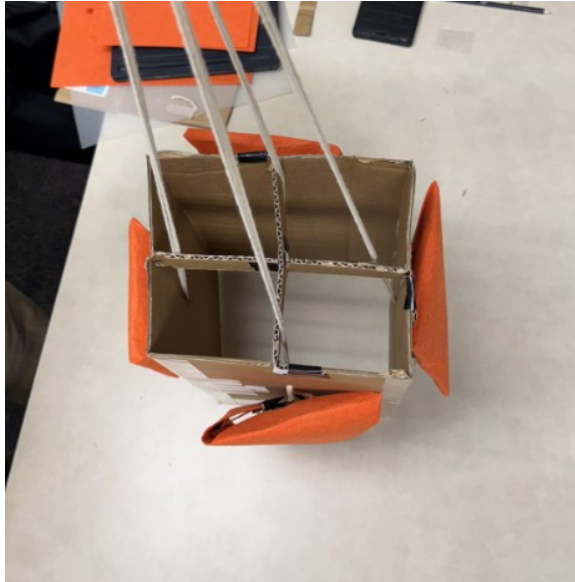
Team Check-In 3: Higher Fidelity Model

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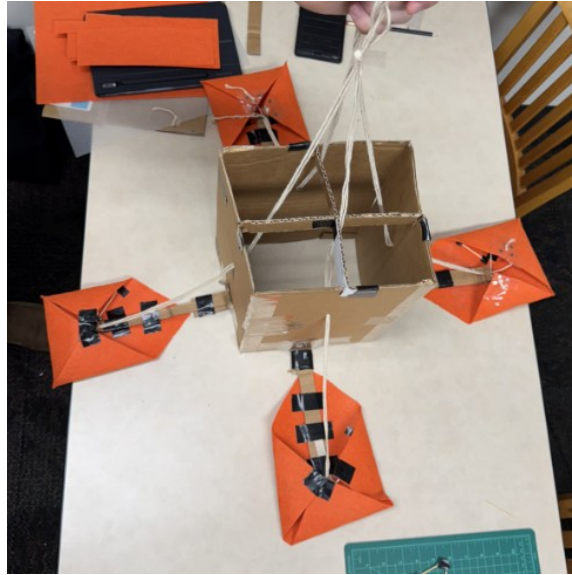
MIME 497 Group 08B



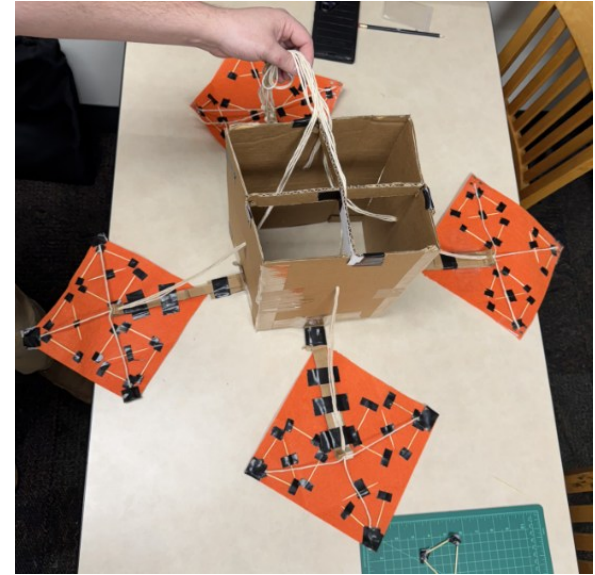
Physical Prototype & Progression



Configuration during transit and decent



Configuration just above surface

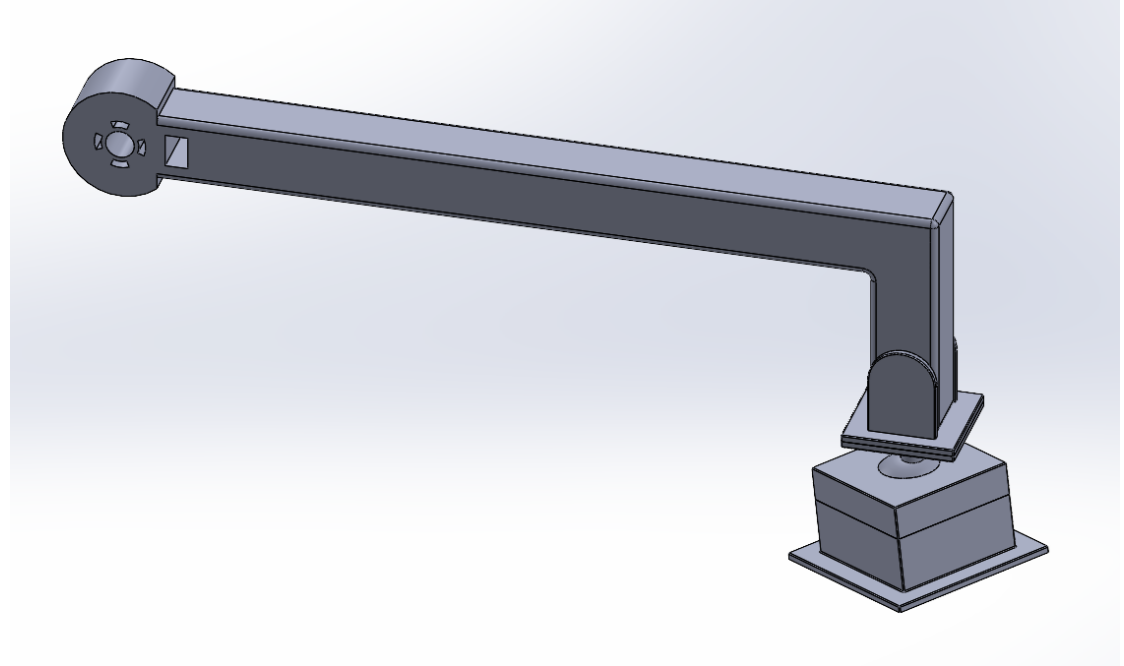


Configuration at surface

CAD Prototype 1: Leg

CAD Key Functions:

- Ball Hinge
- Simple Clutch

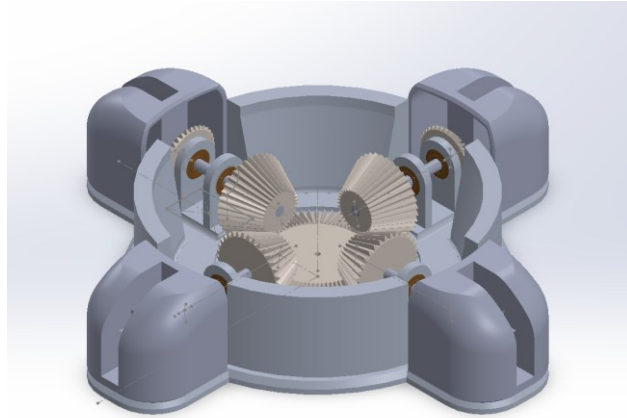


Leg Module of Attachment

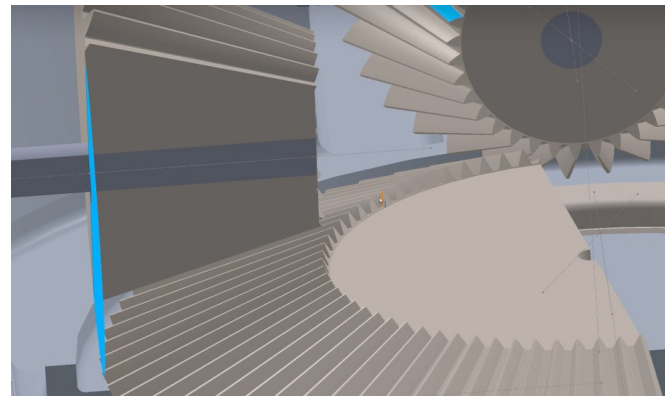
CAD Prototype 2: Black Box

CAD Key Functions:

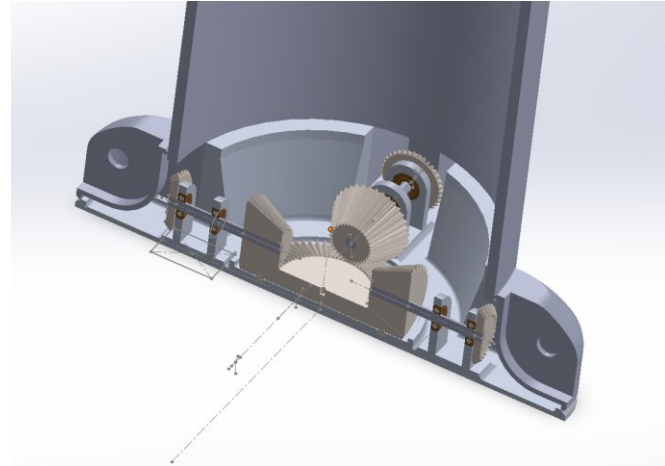
- Single Motor
- Switch that independently stops leg from moving



Black Box Module of Attachment

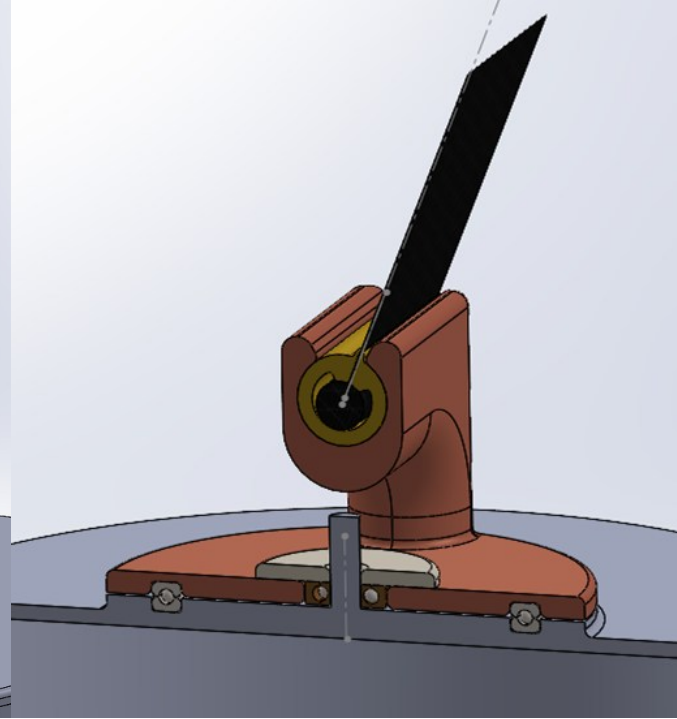
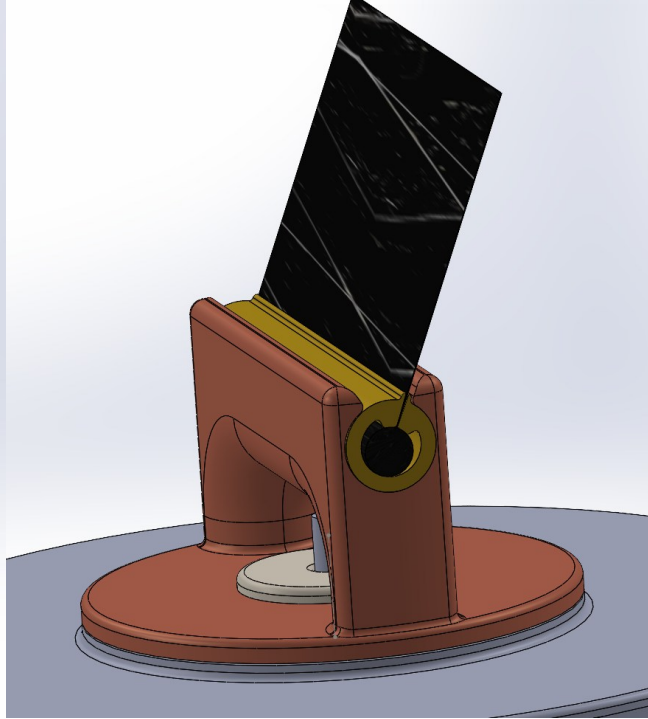
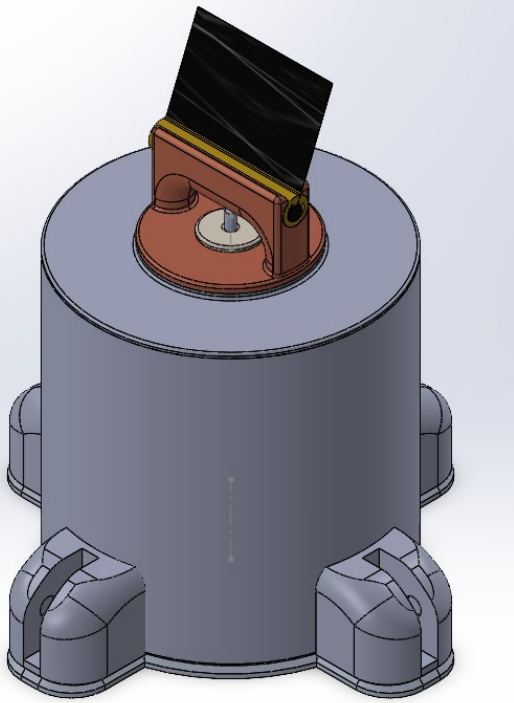


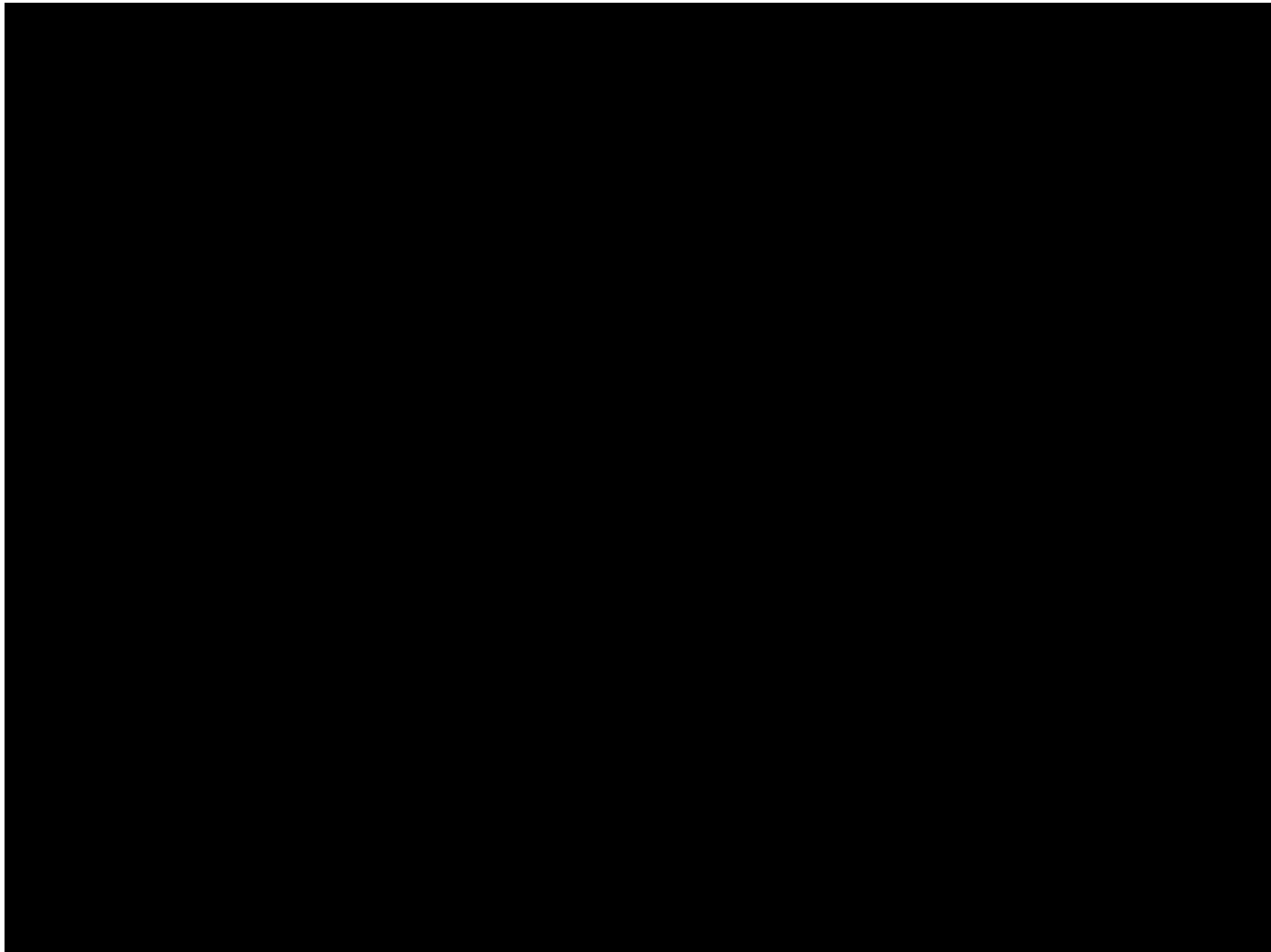
Gear Functionality Exploded View



Cross-Sectional View of Black Box Module

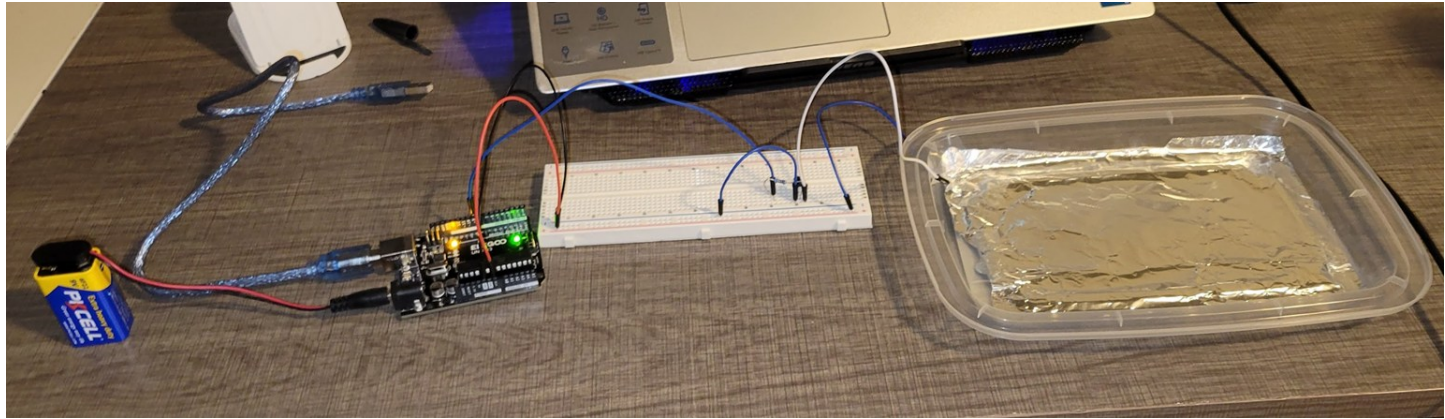
CAD Prototype 3: Tether Attachment Gimbal





Electrostatic Adhesion Demonstration Setup

- Simple circuit involving an Arduino kit, 9V battery, and some aluminum foil with a plastic insulator
- Unable to get a working demonstration at this time, but we understand what is needed to create a working system



Prototype made at home, powered by 9V battery, uses a transistor and 1 k Ω resistor. The 9V powers the bottom of the “tray”, the top of the “tray” is connected to the collector pin on the transistor. The plastic insulator separates the top and bottom pieces of aluminum foil to not short the circuit.

Electrostatic Adhesion Mathematical Proofs

Maxwell Stress Tensor

$$F_{\text{adhesion}} = \frac{\epsilon_0 \epsilon_r A_{\text{real}} V^2}{2h_{\text{eff}}^2} \cdot \beta(\theta, \phi)$$

Roughness Correction

$$F_{\text{rough}} = F_{\text{smooth}} \cdot \frac{A_{\text{real}}}{A_{\text{nominal}}}$$

For a 0.1 m² footpad on a ferrous asteroid ($\epsilon_{\text{surface}} = 8$, $h_{\text{rms}} = 10 \mu\text{m}$):

- $V = 15 \text{ kV}$, $\theta = 40^\circ$, $\phi = 0.2$
- $A_{\text{real}} = 0.032 \text{ m}^2$, $h_{\text{eff}} = 12 \mu\text{m}$, $\beta = 0.85$

$$F_{\text{adhesion}} = \frac{(8.85 \times 10^{-12})(4.5)(0.032)(15,000)^2}{2(12 \times 10^{-6})^2} \cdot 0.85 = 51.3 \text{ N}$$

S. He and R. B. Mrad, "Design, Modeling, and Demonstration of a MEMS Repulsive-Force Out-of-Plane Electrostatic Micro Actuator," Journal of Microelectromechanical Systems, vol. 17, no. 3, pp. 532-541, June 2008

T. Bryan, T. Macleod, L. Gagliano, S. Williams, and B. McCoy, "Innovative Electrostatic Adhesion Technologies," NASA/Langley Research Center, Hampton, 2015

Conclusion

● **Project Successes & Learning**

- Our initial CAD phase validated key design principles, particularly in initial joining & gearing systems, design principles, and CAD proof of concept.

● **Key Takeaways:**

- Interlocking multiplanar gear system in black box
- Flexible leg design with ball joint for improved range of motion
- Tether attachment gimbal
- Electrostatic adhesion progress

● **Next Steps:**

- Iterative design improvements (Gear Design, Feet Design, Motors, Control System/Wiring)
- Move towards advanced 3D prints after next iteration
- Research of material selection, stress/strain calculations, Thermite Welding & Electromagnetic Adhesion proofs & calculations, controls, and CFD

Thank you! Questions?